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### Three-dimensional tortuous path flow element for ball control valves

#### Abstract

A fluid control valve includes a valve housing having a fluid inlet and a fluid outlet. A valve body is disposed within the valve housing and is transitional relative to the valve housing between an open position and a closed position. In the closed position, the valve body prevents fluid flow between the fluid inlet and the fluid outlet. The valve body is incrementally transitional from the closed position to the open position to incrementally increase the amount of fluid flow from the fluid inlet to the fluid outlet. A flow control element is positioned within the valve housing downstream of the valve body. The flow control element includes a peripheral body and an inward body having a plurality of flow control passageways extending therethrough. The flow control element additionally includes an opening defined collectively by the peripheral body and the inward body.

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## **Background/Summary**

### CROSS-REFERENCE TO RELATED APPLICATIONS

(1) Not Applicable

STATEMENT RE: FEDERALLY SPONSORED RESEARCH/DEVELOPMENT

(2) Not Applicable

### BACKGROUND

#### 1. Technical Field

(3) The present disclosure relates generally to a flow control element, and more specifically to a flow control element having a plurality of flow control passageways formed therein and specifically configured for use with a ball-type valve body.

#### 2. Description of the Related Art

(4) Linear fluid control valve assemblies are known and are commonly outfitted to include a noise attenuation or impedance assembly. Such valves are often referred to in the relevant industry as drag valves. Prior art linear valves may include an annular impedance assembly which includes a plurality of annular disks, each defining a plurality of radially extending, tortuous flow passages, with the disks being secured to each other in a stacked arrangement. A piston may be disposed within the interior of the impedance assembly and may be cooperatively engaged to an actuator operative to facilitate the reciprocal movement of the piston within the impedance assembly. When the piston is in a lowermost position, none of the passages of the impedance assembly may be exposed to an incoming flow. However, as the piston is moved upwardly toward an open position, flow passes through the passages of the impedance assembly to provide an exit flow through the linear valve. The amount of flow through the impedance assembly may be varied by the position of the piston, which in turn varies the area or proportion of the impedance assembly exposed to the

incoming flow within the interior thereof.

(5) Though the above-described linear valve arrangement provides significant noise reduction capabilities, in certain applications it is often desirable to employ the use of a rotary valve utilizing a rotary closure element as an alternative to a linear valve.

(6) Accordingly, there is a need in the art for a flow control element that can achieve desired flow characteristics in a rotary valve utilizing a rotary closure element. Various aspects of the present disclosure address this particular need, as will be discussed in more detail below.

#### BRIEF SUMMARY

(7) In accordance with one embodiment of the present disclosure, there is provided a fluid control valve comprising a valve housing having a fluid inlet and a fluid outlet. A valve body is disposed within the valve housing and is transitional relative to the valve housing between an open position and a closed position. In the closed position, the valve body prevents fluid flow between the fluid inlet and the fluid outlet. The valve body is incrementally transitional from the closed position to the open position to incrementally increase the amount of fluid flow from the fluid inlet to the fluid outlet. A flow control element is positioned within the valve housing downstream of the valve body. The flow control element includes a peripheral body and an inward body having a plurality of flow control passageways extending therethrough. The flow control element additionally includes an opening defined collectively by the peripheral body and the inward body.

(8) The valve body and the flow control element may be positioned relative to each other such that as the valve body transitions from the closed position toward the open position, fluid flows through at least one of the plurality of flow control passageways prior to fluid flow through the opening.

(9) The valve housing may define a flow axis extending between the fluid inlet and the fluid outlet. At least one of the plurality of flow control passageways may be comprised of a plurality of interconnected segments, with each segment being inclined 45 degrees relative to the flow axis. At least one of the plurality of flow control passageways may include a rectangular cross section in a plane perpendicular to the flow axis. At least one of the plurality of flow control passageways may include a first rectangular cross section in a first plane perpendicular to the flow axis and a second rectangular cross section in a second plane perpendicular to the flow axis, with the second plane being downstream from the first plane and the second rectangular cross section being larger than the first rectangular cross section.

(10) The flow control element may include a concave surface, and the valve body may include a spherical surface complementary to the concave surface of the flow control element.

(11) The peripheral body may define a central axis and each of the plurality of flow control passageways may extend in a direction parallel to the central axis to define a respective length. The lengths of at least two of the plurality of flow control passageways may be different.

(12) The flow control element may be formed as a unitary structure.

(13) According to another aspect of the present disclosure, there is provided a flow control element for use in a fluid control valve. The flow control element includes a peripheral body and an inward body having a concave surface and a plurality of flow control passageways extending into the inward body from the concave surface. The flow control element additionally includes an opening defined collectively by the peripheral body and the inward body.

(14) According to another aspect of the present disclosure, there is provided a fluid control valve comprising a valve housing having a fluid inlet and a fluid outlet. A valve body is disposed within the valve housing and is transitional relative to the valve housing between an open position and a closed position. In the closed position, the valve body prevents fluid flow between the fluid inlet and the fluid outlet. The valve body may be incrementally transitional from the closed position to the open position to incrementally increase the amount of fluid flow from the fluid inlet to the fluid outlet. A flow control element is positioned within the valve housing downstream of the valve body. The flow control element includes an orifice plate having a plurality of passage inlets formed therein and at least one flow opening formed therein. The flow control element additionally

includes a main body having a plurality of flow control passageways extending therethrough. The main body is aligned with the orifice plate such that the plurality of passage inlets are in communication with a respective one of the flow control passageways.

(15) The flow control element may additionally include a frame engaged with the orifice plate and the main body.

(16) The main body may include an inlet face and an outlet face spaced from the inlet face along a main body axis. At least one of the plurality of flow control passageways may be comprised of a plurality of interconnected segments, with each segment being inclined 45 degrees relative to the main body axis.

(17) The orifice plate may include a concave surface and the valve body may include a spherical surface complementary to the concave surface of the orifice plate.

(18) The present disclosure will be best understood by reference to the following detailed description when read in conjunction with the accompanying drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features and advantages of the various embodiments disclosed herein will be better understood with respect to the following description and drawings, in which:

(2) FIG. 1 is a cross-sectional view of a fluid control valve having a ball-type valve body and a downstream flow control element adjacent the ball-type valve body;

(3) FIG. 1A is an upper perspective, partial cross-sectional view of another embodiment of a fluid control valve having a ball-type valve body and a downstream flow control element adjacent the ball-type valve body;

(4) FIG. 2 is an upper perspective view of the flow control element incorporated into the fluid control valve shown in FIG. 1A;

(5) FIG. 3 is a front view of the flow control element depicted in FIG. 2;

(6) FIG. 4 is a cross-sectional view of the flow control element depicted in FIG. 3;

(7) FIG. 5 is a lower front perspective view of the flow control element, with a plurality of internal flow control passageways thereof being depicted in phantom;

(8) FIG. 6 is a perspective view of an exemplary flow control passageway formed in the flow control element;

(9) FIG. 7 is an upper perspective view of one embodiment of a flow control element;

(10) FIG. 8 is a front view of the flow control element depicted in FIG. 7;

(11) FIG. 9 is an upper perspective view of another embodiment of a flow control element;

(12) FIG. 10 is a front view of the flow control element depicted in FIG. 9;

(13) FIG. 11 is an upper perspective view of yet another embodiment of a flow control element;

(14) FIG. 12 is a front view of the flow control element depicted in FIG. 11;

(15) FIG. 13 is an upper perspective view of a further embodiment of a flow control element;

(16) FIG. 14 is a front view of the flow control element depicted in FIG. 13;

(17) FIG. 15 is an exploded upper perspective view of a multi-component flow control element;

(18) FIG. 16 is an end view of an exemplary flow control passageway which may be included in the multi-component flow control element of FIG. 15;

(19) FIG. 17 is an upper perspective view of the flow control passageway depicted in FIG. 16;

(20) FIG. 18 is a front view of the two-stage flow control passageway depicted in FIG. 17;

(21) FIG. 19 is a front view of an exemplary four-stage flow control passageway which may be included in the multi-component flow control element of FIG. 15;

(22) FIG. 20 is a front view of an exemplary six-stage flow control passageway which may be included in the multi-component flow control element of FIG. 15;

- (23) FIG. **21** is an upper perspective, partial cross-sectional view of another embodiment of a fluid control valve having a ball-type valve body and a flow control element integrated into ball-type valve body;
- (24) FIG. **22** is a front view of the ball-type valve body having the integrated flow control element;
- (25) FIG. **23** is a cross-sectional view of the ball-type valve body and integrated flow control element of FIG. **22**; and
- (26) FIG. **24** is an upper perspective view of the flow control element shown in FIGS. **21-23**.
- (27) Common reference numerals are used throughout the drawings and the detailed description to indicate the same elements.

#### DETAILED DESCRIPTION

(28) The detailed description set forth below in connection with the appended drawings is intended as a description of certain embodiments of a flow control valve and is not intended to represent the only forms that may be developed or utilized. The description sets forth the various structure and/or functions in connection with the illustrated embodiments, but it is to be understood, however, that the same or equivalent structure and/or functions may be accomplished by different embodiments that are also intended to be encompassed within the scope of the present disclosure. It is further understood that the use of relational terms such as first and second, and the like are used solely to distinguish one entity from another without necessarily requiring or implying any actual such relationship or order between such entities.

(29) Referring now to FIGS. **1** and **1A**, there is provided a flow control valve **10** having a flow control element **12** and a generally spherical, ball-type valve body **14** or similar rotary closure valve body. The flow control element **12** may be positioned adjacent the valve body **14** and include a surface that may be complementary in shape to the valve body **14**. The flow control element **12** may include two regions, including a restricted region having a plurality of tortuous path stages formed therein, and an unrestricted region including an opening to allow for unrestricted flow therethrough. The flow control element **12** may be positioned relative to the valve body **14** such that as the valve body **14** begins to move from a closed position toward an open position, fluid flows initially through the restricted region before fluid flows through the unrestricted region. The flow control element **12** may be formed from additive manufacturing methods (e.g., three-dimensional printing), to allow for an increased flow capacity and/or the number of tortuous path stages therein. In this regard, the flow control valve **10** allows for desired flow control during operation of the valve body **14**.

(30) The flow control element **12** and valve body **14** may be disposed within a valve housing **16** having a fluid inlet **18**, a fluid outlet **20**, and an internal flow path **22** extending therebetween along a flow axis **24**. The valve housing **16** additionally includes an internal chamber **26**, which houses the valve body **14** and the flow control element **12**. As noted above, various aspects of the present disclosure are directed toward the use of a ball-type valve body **14**, and thus, the internal chamber **26** may be sized to allow the valve body **14** to rotate about rotation axis **28**, which may be perpendicular to the flow axis **24**. An actuation stem **30** may be connected to the valve body **14** and may be used control movement of the valve body **14** between the closed and open positions.

(31) The valve body **14** may include a valve flow passage **32** extending therethrough between an inlet opening and an outlet opening. The valve body **14** is transitional relative to the valve housing **16** between an open position and a closed position. In the open position, the inlet opening of the valve body **14** is in fluid communication with the fluid inlet **18** of the valve housing **16**, and the outlet opening of the valve body **14** is in fluid communication with the fluid outlet **20** of the valve housing **16**. Accordingly, when the valve body **14** is in the open position, fluid may flow from the fluid inlet **18** of the valve housing **16**, through the valve body **14**, and into the fluid outlet **20** of the valve housing **16**. In the closed position, the valve flow passage **32** is moved out of fluid communication with the fluid inlet **18** and fluid outlet **20** of the valve housing **16** so as to prevent fluid flow through the valve body **14**. In particular, side walls of the valve body **14** may be moved

into alignment with the fluid inlet **18** and fluid outlet **20** of the valve housing **16** to prevent fluid flow through the valve body **14** when the valve body **14** is in the closed position. In one embodiment, the valve body **14** may rotate approximately 90 degrees between the open position and the closed position.

(32) According to one embodiment, the valve body **14** may be incrementally transitional between the closed position and the open position to incrementally control the amount of fluid flowing between the fluid inlet **18** to the fluid outlet **20**. In other words, each degree that the valve body **14** moves from the closed position toward the open position may allow for a greater amount of fluid flow from the fluid inlet **18** to the fluid outlet **20**. Conversely, each degree that the valve body **14** moves from the open position toward the closed position may allow for a smaller amount of fluid flow from the fluid inlet **18** to the fluid outlet **20**.

(33) The flow control element **12** is positioned within the valve housing **16** adjacent the valve body **14**. In the exemplary embodiment, the flow control element **12** is downstream of the valve body **14**, although it is contemplated that the flow control element **12** may also be incorporated into the valve body **14** in other implementations.

(34) The flow control element **12** is shown in more detail in FIGS. 2-4 and includes a peripheral body **34**, an inward body **36**, and a support rib **38** extending between the peripheral body **34** and the inward body **36**. The flow control element **12** may include one or more openings **39** collectively defined by the inward body **36** and the peripheral body **34**. In the exemplary embodiment depicted in FIGS. 2-4, the flow control element **12** includes four large openings **39**, which are separated by the support rib **38**. The inward body **36** may define the restricted region of the flow control element **12**, while the openings **39** may define the unrestricted region of the flow control element **12**, as will be described in more detail below.

(35) The peripheral body **34** of the flow control element **12** may be disposed about a central axis **40** and include a first outer surface **42** and a second outer surface **44**, both of which are circular in configuration, with the second outer surface **44** being smaller in diameter than the first outer surface **42**. A shoulder **46** extends between the first and second outer surfaces **42**, **44**. The first outer surface **42** extends between the shoulder **46** and an end surface **48** opposite the shoulder **46**. A plurality of mount openings **50** may extend through the flow control element **12** between the shoulder **46** and the end surface **48**. The mount openings **50** may be configured to receive a mechanical fastener, such as a bolt, which may be used for mounting the flow control element **12** to the valve housing **16**. The second outer surface **44** may extend from the shoulder **46** and terminate at an outer rim **50**, which may define an outer boundary for fluid flow, as will be described in more detail below.

(36) The inward body **36** extends radially inward from at least a portion of the peripheral body **34**. The inward body **36** includes an upstream surface **54** and a downstream surface **56** opposite the upstream surface **54**. A plurality of flow control passageways **58** may extend through the inward body **36** between the upstream surface **54** and the downstream surface **56**. In the embodiment depicted in FIGS. 2-4, the flow control passageways **58** are arranged in a rectangular grid arrangement, e.g., aligned in rows and columns. The upstream surface **54** may be concave and may be configured to interface with the outer contour of the ball-type valve body **14**. In this regard, at least a portion of the outer surface of the ball-type valve body **14** may be of a spherical configuration (e.g., convex), and thus, the concave configuration of the upstream surface **54** may be complementary to the spherical configuration of the ball-type valve body **14**.

(37) As shown in FIG. 4, the downstream surface **56** may define a stepped configuration, with the spacing between the downstream surface **56** and the end surface **48** varying along the downstream surface **56**. In the embodiment depicted in FIG. 4, the downstream surface **56** includes a first region **57**, a second region **59**, and a third region **61**. From the perspective shown in FIG. 4 and starting at the top of the downstream surface **56**, the first region **57** includes an outwardly angled segment **63** which transitions into a parallel segment **65** that is parallel to the end surface **48**. The distance

between the parallel segment **65** of the first region **57** and the end surface **48** defines a first thickness  $T_{sub.1}$ . The second region **59** includes an inwardly angled segment **67** which transitions into a parallel segment **69** that is parallel to the end surface **48**. The distance between the parallel segment **69** of the second region **59** and the end surface **48** defines a second thickness  $T_{sub.2}$ . The third region **61** includes an inwardly angled segment **71** which transitions into a parallel segment **73** that is parallel to the end surface **48**. The distance between the parallel segment **73** of the third region **61** and the end surface **48** defines a third thickness  $T_{sub.3}$ . The first thickness  $T_{sub.1}$  is greater than the second thickness  $T_{sub.2}$ , which is greater than the third thickness  $T_{sub.3}$ . The variation in thickness may allow for a variation in the length of the passageways extending through the flow control element **12a**. Therefore, a given flow control element **12a** may be capable of inducing a pressure drops that vary in magnitude based on the fluid passageway through which the fluid flows.

(38) It is also contemplated that other embodiments of the flow control element may have a downstream surface that is not stepped. For instance, the downstream surface **56** may have a spherical, convex configuration.

(39) FIG. 5 is a lower perspective view of another embodiment of a flow control element **12** which may include a pair of larger openings **39** and a plurality of flow control passageways **58** arranged in a polar coordinate configuration, e.g., the openings of the flow control passageways **58** are aligned along respective radii extending from central axis **40**.

(40) In a given flow control element **12**, the length of the flow control passageways **58** may be defined by the distance along the passageway **58** between the upstream surface **54** and the downstream surface **56**. With the configuration of the upstream surface **54** being primarily dictated by the outer contour of the valve body **14**, the configuration of the downstream surface **56** may be dictated by a desired length of the flow control passageways **58**. Thus, the downstream surface **56** may be convex, concave, planar, stepped, or other configurations known in the art to achieve a desired length of the flow control passageways **58**.

(41) The flow control passageways **58**, which each extend between the upstream surface **54** and the downstream surface **56**, as indicated above, are designed to impart a pressure drop on the fluid flowing therethrough. Accordingly, each flow control passageway **58** may include a series of bends or turns to define a tortuous configuration. The series of bends or turns may define one or more stages in the passageway **58**, with adjacent stages being separated by a turn or bend. In a given passageway **58**, increasing the number of stages allows for a larger pressure drop to be achieved by the fluid flowing through the passageway **58**.

(42) FIG. 6 is an upper perspective view of a flow control passageway **58** isolated by itself (e.g., now shown in the flow control element **12**) to depict an exemplary configuration thereof. The flow control passageway **58** includes an inlet **60** and an outlet **62**, with the flow control passageway **58** extending between the inlet **60** and the outlet **62** along a passageway axis **64**. According to one embodiment, the flow control passageway **58** may be comprised of a plurality of interconnected segments **66**. In the embodiment depicted in FIG. 6, the flow control passageway **58** includes 14 segments **66a-n**. Each segment **66** may be inclined relative to the passageway axis **64** by an angle. Elements **66a**, **66c**, **66e**, **66g**, **66i**, **66k**, **66m** extend in a direction parallel to axis **68**, while elements **66b**, **66d**, **66f**, **66h**, **66j**, **66l**, **66n** extend in a direction parallel to axis **70**. Axis **68** is angled relative to the passageway axis **64** by an angle  $\Theta_{sub.1}$ , while axis **70** is angled relative to the passageway axis **64** by an angle  $\Theta_{sub.2}$ . The magnitudes of  $\Theta_{sub.1}$  and  $\Theta_{sub.2}$  may vary between 0 and 90 degrees, although in the embodiment depicted in FIG. 6, the magnitude of  $\Theta_{sub.1}$  and  $\Theta_{sub.2}$  is approximately 45 degrees. Thus, each segment **66** is inclined 45 degrees relative to the passageway axis **64**.

(43) The flow control passageways **58** may include a rectangular cross section in a plane perpendicular to the passageway axis **64**. Furthermore, the cross-sectional size of the flow control passageways **58** may increase along the length of the flow control passageway **58** from the inlet **60**

to the outlet **62**. In other words, the flow control passageway **58** may define a first rectangular cross section in a first plane perpendicular to the passageway axis **64** and a second rectangular cross section in a second plane perpendicular to the passageway axis **64**, with the second plane being downstream from the first plane and the second rectangular cross section being larger than the first rectangular cross section.

(44) The inward body **36** may be configured such that the lengths of the flow control passageways **58** may vary. For instance, the lengths of the flow control passageways **58** that are a common radial distance from the central axis **40** may be equal. Thus, the flow control passageways **58** positioned closer to the central axis **40** may be shorter in length to the flow control passageways **58** positioned farther away from the central axis **40**. An example of this configuration is shown in FIG. **4** which shows a group of six-stage flow control passageways **58** furthest away from the central axis **40**, then a group of three-stage flow control passageways **58** closer to the central axis **40**, and then a group of two-stage passageways **58** closest to the central axis **40**. In an alternative embodiment, the length of the passageways **58** may vary along a prescribed diameter of the peripheral body **34**. For instance, the flow control passageways **58** may be longer at the bottom of the flow control element **12** and shorter at the top, or vice versa, or alternatively, the flow control passageways **58** may be longer at one side of the flow control element **12** and shorter at the opposing side or vice versa. Along these lines, those of ordinary skill in the art will recognize that the longer the flow control passageway **58**, the greater the ability to potentially increase the number of turns or stages defined thereby (and hence the energy dissipation capacity thereof). In summary, the lengths of the flow control passageways **58** and/or the number of stages defined thereby may vary according to any scheme or pattern as may be desired to effectuate a desired flow characteristic, e.g., a desired pressure drop in fluid flowing therethrough. Exemplary arrangements of the flow control passageways **58** in the context of a presently contemplated functional characteristic of the flow control element **12** within the flow control valve **10** will be discussed in more detail below.

(45) Referring now to FIGS. **7-14**, there is depicted several different embodiments of the flow control element **12**, with the primary distinction between the embodiments being the sizes of the inward body **36** (e.g., the restricted region) and the openings **39** (e.g., the unrestricted region) extending within the peripheral body **34**. In this regard, the relative surface area percentages of the flow control element **12** assumed by the inward body **36** and the opening(s) **39** may vary depending on the desired flow characteristics. FIGS. **7** and **8** show a first embodiment of the flow control element **12a**, wherein the inward body **36a** is the smallest of those flow control elements **12a-d** depicted in FIGS. **7-14**, while the collective sizes of the openings **39a** is the largest of those depicted in FIG. **7-14**. More specifically, the inward body **36a** constitutes less than half of the area defined by the outer rim **52**, while the openings **39a** collectively constitute more than half of the area defined by the outer rim **52**.

(46) FIGS. **9** and **10** depict a second embodiment of the flow control element **12b**, wherein the inward body **36b** constitutes approximately half of the area defined by the outer rim **52**, and the openings **39b** collectively constitute approximately half of the area defined by the outer rim **52**.

(47) FIGS. **11** and **12** depict a third embodiment of the flow control element **12c**, wherein the inward body **36c** constitutes more than half of the area defined by the outer rim **52**, and the openings **39c** collectively constitute less than half of the area defined by the outer rim **52**.

(48) FIGS. **13** and **14** depict a fourth embodiment of the flow control element **12d**, wherein the inward body **36d** extends across the entirety of the area defined by the outer rim **52**, thus leaving no remaining area for any openings **39**.

(49) When placed in the valve housing **16**, the flow control element **12** may be positioned relative to the valve body **14** such that as the valve body **14** transitions from the closed position toward the open position, fluid flows through at least one of the plurality of flow control passageways **58** prior to fluid flow through the openings **39**. Accordingly, as the valve body **14** transitions from the closed position toward the open position, the valve flow passage **32** of the valve body is first brought into

alignment with the inward body **36**, while the openings **39** remain out of alignment with the valve flow passage **32** of the valve body **14**. Accordingly, fluid may flow through at least some of the flow passageways **58**, while fluid flow through the openings **39** remains restricted. It is contemplated that when the valve body **14** is only slightly moved from the closed position toward the open position, only a small percentage of the passageways **58** may be in communication with the valve flow passage **32**, and thus, only a small percentage of the passageways **58** may have fluid flowing therethrough.

(50) As the valve body **14** continues toward the open position, a greater percentage of the inward body **36** may be brought into alignment with the valve flow passage **32**, which in turn, may increase the number of flow control passageways **58** receiving fluid therefrom.

(51) As the valve body **14** further continues toward the open position, the openings **39** (if included) may be brought into alignment with the valve flow passage **32**. When such alignment occurs, fluid may flow through the openings **39**, while also flowing through the flow control passageways **58**. The openings **39** allow for unimpeded fluid flow therethrough, whereas the flow control passageways **58** provide resistance in the form of the multi-stage configuration thereof. Accordingly, turndown and noise performance of the ball-type flow control valve **10** may be improved in various flow control applications.

(52) The configuration of the flow control element **12** may be designed to control or regulate the fluid pressure/velocity when the valve body **14** opens. Oftentimes, the pressure/velocity of the fluid at the initial opening of the valve body **14** is of greater magnitude (and greater concern), than the pressure/velocity of the fluid when the valve body **14** is completely open. Thus, the flow control passageways **58** may be configured and arranged to provide a greater degree of energy attenuation immediately at the opening of the valve body **14**, with those energy attenuating attributes decreasing as the valve body **14** continues to transition to its fully open position, eventually culminating in at least a portion of the flow control element **12** (e.g., those portions defined by the openings **39**) providing virtually no flow restriction when the valve body **14** is completely open.

(53) By way of example, and using the perspectives shown in FIGS. **8**, **10**, **12**, and **14** the right-most region of the flow control element **12** may receive the fluid at greatest pressure/velocity as the fluid exits the valve body **14** at initial opening. Thus, those flow control passageways **58** within this region (considering the structural attributes of the flow control element **12** shown and described above in relation to FIG. **4**) may be greater in length and may further define the greatest number of stages (and hence energy attenuating capability). As valve flow passage **32** of the valve body **14** moves across the flow control element **12** in the direction of arrow **75**, which occurs as the valve body **14** gradually transitions toward its fully open position, the flow control passageways **58** arranged inwardly from the right-most region toward the axis **24** will progressively be exposed to fluid flow. As it is contemplated that less energy attenuating capacity may be needed when the valve body **14** is opened to the corresponding degree, these particular flow controls passageways **58**, which are typically slightly shorter in length and/or have fewer stages than those that are located within the right-most region, will thus provide a comparatively reduced level of energy attenuation. The trend of shorter flow control passageways **58** and/or fewer stages (and hence reduced energy attenuation capability) may continue as the valve flow passage **32** continues along the direction of arrow **75**. Eventually, as the valve body **14** approaches its fully open position, its valve flow passage **32** may communicate with the opening(s) **39** (if included) in the flow control element **12** to allow for unimpeded fluid flow with virtually no energy attenuation being facilitated by such opening(s).

(54) With regard to the embodiment depicted in FIGS. **13** and **14**, while the flow control element **12d** may be formed without any one or more openings **39**, the flow control passageways **58** positioned adjacent the left-most region, may be formed to be of the shortest in length, or may have no turns despite being of the greatest length, to comparatively reduce their energy attenuating capability. Along these lines, fluid flow through such left-most flow control passageways **58** may

only occur when the valve body **14** is completely open, and thus, the energy attenuating needs for these particular flow control passageways **58** may be lower than for those flow control passageways **58** exposed to fluid flow at the initial opening of the valve body **14**.

(55) The unique configuration of the flow control element **12**, and in particular, the complex configuration of the flow control passageways **58**, may be made possible by additive manufacturing or three-dimensional printing, wherein the flow control element **12** is formed in successive layers to form a unitary structure. Conventional flow control elements, particularly those used in connection with ball-type valve bodies, typically included a series of stacked plates to define flow control passageways. However, the stacked plate arrangement was associated with inherent limitations as to the complexity of the flow control passageways defined thereby. Accordingly, the ability to create complex flow control passageways **58** with accuracy and precision via additive manufacturing presents a significant improvement over conventional stacked plate flow control elements. An example of an additive manufacturing technique is direct metal laser sintering (DMLS), which is described in U.S. Pat. No. 8,826,938 entitled Direct Metal Laser Sintered Flow Control Element, the disclosure of which is incorporated herein by reference. Other three-dimensional printing or manufacturing techniques known in the art may also be used without departing from the spirit and scope of the present disclosure.

(56) For larger flow control elements, three-dimensional printing of the flow control element as a single unitary structure may not be feasible, and thus, alternative embodiments of the flow control element may include a multi-component structure. An example of a multi-component flow control element **70** is depicted in FIG. **15** and generally includes an orifice plate **72**, at least one main body **74**, and a frame **76**, with the flow control element **70** being disposed about a central axis **77**. The orifice plate **72** includes first outer surface **78**, a second outer surface **80**, and an upstream surface **82** having a plurality of passage inlets **84** formed therein. Both the first and second outer surfaces **78**, **80** may be circular in configuration, with the second outer surface **80** being smaller in diameter than the first outer surface **78**. A shoulder **86** may extend between the first and second outer surfaces **78**, **80**, with the shoulder **86** including a plurality of mount openings **88** formed therein. The second outer surface **80** may extend from the shoulder **86** and terminate at an outer rim **90**, which may define an outer boundary for fluid flow. The upstream surface **82** may be concave in configuration so as to interface with the ball-type valve body **14** having the spherical outer surface configuration. The upstream surface **82** and the second surface **80** may collectively define a flow opening **92** which allows for unrestricted fluid flow therethrough. One or more support ribs may extend between the second surface **80** and the upstream surface **82** and may separate adjacent flow openings.

(57) The flow control element **70** shown in FIG. **15** includes two main bodies **74**, each of which include a plurality of flow control passages **94** extending therethrough. In this regard, each main body **74** includes a first surface **96** and a second surface **98** opposite the first surface **96**, with the flow control passages **94** extending from the first surface **96** to the second surface **98**. Each main body **74** may be aligned relative to the orifice plate **72** such that the plurality of passage inlets **84** of the orifice plate **72** may be in fluid communication with a respective one of the flow control passages **94**.

(58) The frame **76** may support the main bodies **74** and the orifice plate **72**. The frame **76** may include an outer wall **100** and a flange **102** extending radially outward from the outer wall **100**. The frame **76** may additionally include one or more support walls **104** for defining body support openings **106** sized to receive respective ones of the main bodies **74**. The number of body support openings **106** may be equal to the number of main bodies **74**. Although the exemplary embodiment shows two body support openings **106** and two main bodies **74**, it is understood that some embodiments may include only one body support opening **106** and one main body **74**, while other embodiments may include more than two body support openings **106** and more than two main bodies **74**. The support walls **104** and the outer wall **100** may also collectively define one or more

flow openings **108**, which may be placed in communication with the flow openings **92** on the orifice plate **72**. The flange **102** may include a plurality of mount openings **111** extending therethrough, and alignable with the mount openings **88** on the orifice plate **72** to facilitate assembly of the flow control element **70** and attachment of the flow control element **70** to the valve housing **16**. Assembly of the flow control element **70** may entail placement of the main bodies **74** in the body support openings **106** of the frame **76** and placement of the orifice plate **72** over the frame **76**. In this regard, the first outer surface **78** of the orifice plate **72** may be placed over the flange, while the shoulder **86** on the orifice plate **72** may be placed in contact with the flange **102**. (59) With further regard to FIG. **15**, a variant of the flow control element **70** is contemplated wherein the orifice plate **72** is eliminated from the assembly. In greater detail, it is envisioned that the main bodies **74** may be fabricated in a manner wherein, when integrated into the frame **76**, the first surfaces **96** thereof collectively define an arcuately contoured, generally concave surface, the profile of which generally approximates that portion of the upstream surface **82** of the orifice plate **72** including the passage inlets **84** therein. In other words, when the main bodies **74** are mechanically retained within the frame **76**, the convex surface collectively defined thereby may directly interface with the concave surface of the ball **14**.

(60) Referring now to FIGS. **16-20**, there is shown various views of exemplary flow control passageways **113** that may be incorporated into the flow control element **70**. Each flow control passageway **113** may be collectively defined by a passage inlet **84** on the orifice plate **72** and the flow control passages **94** in the main bodies **74**. As can be seen, the flow control passageways **113** may include a main portion that defines a rectangular cross-sectional shape, and which may increase in size along the length of the flow control passageway **113** from the inlet to the outlet. FIG. **18** shows a two-stage passageway **112**, FIG. **19** shows a four-stage passageway **113**, and FIG. **20** shows a six-stage passageway **113**. Furthermore, each of the flow control passageways **113** may additionally include a geometric loft **115** at the inlet end portion thereof. The loft **115** may provide a smooth transition between a round inlet opening and a rectangular flow path. The inlet end portion of the loft **115** may also include one or more inlet splitter plates **116**. The splitter plates **116** may extend diametrically across the inlet opening and prevent the passage and collection of large particulate into the throat of the geometric loft **115**.

(61) Those of ordinary skill in the art will recognize that the various structural features of the flow control element **70** may be formed and assembled such that the resultant flow control element **70** mirrors the progressively reduced energy attenuating characteristics of the flow control element **12** as described above as valve body **14** gradually transitions from its fully closed position to its fully open position.

(62) Although the foregoing describes the various flow control elements as being separate from the valve body **14**, it is contemplated that in certain embodiments, a flow control element may be mounted directly to the valve body **14** or formed in conjunction with the valve body **14**. FIGS. **21-24** depict an example of a flow control valve **110** having a flow control element **112** and a generally spherical, ball-type valve body **114**. In this regard, when mounted to, or formed as part of the valve body **114**, the flow control element **112** moves with the valve body **114** between its closed and open positions. Furthermore, incremental movement of the valve body **114** between the closed and open positions may allow for incremental exposure of the flow control passageways formed in the flow control element **112** to fluid flow.

(63) FIGS. **22** and **23** show the flow control element **112** integrated into the valve body **114**, while FIG. **24** shows the flow control element **112** by itself. The flow control element **112** shares several structural attributes with the flow control element **12** discussed above. More specifically, the flow control element **112** includes a peripheral body **134**, an inward body **136**, and a support rib **138** extending between the peripheral body **134** and the inward body **136**. The flow control element **112** may include one or more openings **139** collectively defined by the inward body **136** and the peripheral body **134**. A plurality of flow control passageways **158** may extend through the inward

body **136** between an upstream surface **154** and a downstream surface **156**.

(64) A unique feature of the in-ball flow control element **112** is the skirt or extended annular wall **150**, which allows for attachment of the flow control element **112** to the valve body **114**. In particular, the annular wall **150** may include an outer surface defining an outer diameter that is substantially equal to, yet slightly smaller than the inner diameter defined by the valve body **114**. In this regard, the annular wall **150** may be configured to provide an interference fit with the valve body **114**, or alternatively, mechanical fasteners may be used to secure the flow control element **112** to the valve body **114**.

(65) The flow control element **112** may be coupled to an inlet side of the valve body **114**, such that fluid entering the valve body **114** initially passes through the flow control element **112** and then continues through the valve body **114**. From the perspective shown in FIG. **21**, the valve body **114** is shown in a closed position, and rotates in the direction of arrow **125** to transition from the closed position toward the open position. The flow control element **112** may be configured such that the longer flow control passageways **158** are first to receive fluid as the valve body **114** transitions from a closed position toward an open position, and then shorter flow control passageways **158** are subsequently exposed to the fluid flow as the valve body **114** continues toward the open position.

(66) Although FIGS. **1** and **1A** show a flow control element **12** positioned downstream of the valve body **14**, and FIG. **21** shows a flow control element **112** integrated into a valve body **114**, it is contemplated that certain embodiments may essentially include a combination of what is shown in FIGS. **1-1A** and FIG. **21**. In particular, a flow control valve may include a first flow control element integrated into a valve body, and a second flow control element positioned downstream of the valve body.

(67) The particulars shown herein are by way of example only for purposes of illustrative discussion and are not presented in the cause of providing what is believed to be most useful and readily understood description of the principles and conceptual aspects of the various embodiments of the present disclosure. In this regard, no attempt is made to show any more detail than is necessary for a fundamental understanding of the different features of the various embodiments, the description taken with the drawings making apparent to those skilled in the art how these may be implemented in practice.

## Claims

1. A fluid control valve comprising: a valve housing having a fluid inlet and a fluid outlet; a flow axis connecting the fluid inlet and the fluid outlet; a valve body disposed within the valve housing and transitional relative to the valve housing between a fully open position and a closed position, in the closed position, the valve body preventing fluid flow between the fluid inlet and the fluid outlet, the valve body being incrementally transitional from the closed position to the fully open position to incrementally increase the amount of fluid flow from the fluid inlet to the fluid outlet; and a flow control element positioned within the valve housing downstream of the valve body, the flow control element having: a peripheral body defining an end surface and bisected by the flow axis; a single-piece inward body at least partially surrounded by the peripheral body such that a common axis extending perpendicularly from the flow axis extends through portions of the inward body and the peripheral body, the inward body having a concave upstream surface which is complementary to an exterior profile of the valve body, a downstream surface opposite the upstream surface, and a plurality of multi-stage flow control passageways extending between the upstream and downstream surfaces, wherein adjacent stages in each of the multi-stage flow control passageways are separated by a turn or bend, the downstream surface having a stepped configuration including at least two regions which each define a surface segment which is generally parallel to the end surface, at least some of the multi-stage flow control passageways extending through the inward body between the upstream surface and the surface segment of each of the at

least two regions; and at least one opening collectively defined by the peripheral body and the inward body.

2. The fluid control valve recited in claim 1, wherein the valve body and flow control element are positioned relative to each other such that as the valve body transitions from the closed position toward the open position, fluid flows through at least one of the plurality of flow control passageways prior to fluid flow through the opening.

3. The fluid control valve recited in claim 1, wherein at each of the plurality of multi-stage flow control passageways is comprised of a plurality of segments, each segment being inclined 45 degrees relative to the flow axis.

4. The fluid control valve recited in claim 3, wherein each of the plurality of multi-stage flow control passageways includes a rectangular cross section in a plane perpendicular to the flow axis.

5. The fluid control valve recited in claim 3, wherein each of the plurality of multi-stage flow control passageways includes a first rectangular cross section in a first plane perpendicular to the flow axis and a second rectangular cross section in a second plane perpendicular to the flow axis, the second plane being downstream from the first plane and the second rectangular cross section being larger than the first rectangular cross section.

6. The fluid control valve recited in claim 1, wherein the downstream surface of the inward body of the flow control element includes three regions which each define a surface segment which is generally parallel to the end surface.

7. The fluid control valve recited in claim 1, wherein the multi-stage flow control passageways are formed to provide varying degrees of energy attenuating capacity, and are arranged so that those multi-stage flow control passageways exposed to fluid flow upon the initial opening of the valve body provide greater energy attenuating capacity than those multi-stage flow control passageways progressively exposed to fluid flow as the valve body continues to transition to the fully open position.

8. The fluid control valve recited in claim 1, wherein each of the plurality of multi-stage flow control passageways extends along a passageway axis which is parallel to the flow axis to define a respective length, the lengths of at least two of the plurality of multi-stage flow control passageways being different.

9. The fluid control valve recited in claim 1, wherein the flow control element is formed as a unitary structure.

10. A flow control element for use in a fluid control valve having a valve body disposed within a valve housing having a fluid inlet and a fluid outlet connected by a flow axis, and being transitional relative to the valve housing between a closed position and a fully open position, fluid flow through the fluid control valve incrementally increasing as the valve body incrementally transitions from the closed position toward the fully open position, the flow control element comprising: a peripheral body defining an end surface and bisected by the flow axis when the flow control element is disposed within the valve housing; and a single-piece inward body at least partially surrounded by the peripheral body such that when the flow control element is disposed within the valve housing, a common axis extending perpendicularly from the flow axis extends through portions of the inward body and the peripheral body, the inward body having a concave upstream surface which is complementary to an exterior profile of the valve body, a downstream surface opposite the upstream surface, and a plurality of multi-stage flow control passageways extending through the inward body between the upstream and downstream surfaces thereof, wherein adjacent stages in each of the multi-stage flow control passageways are separated by a turn or bend, the downstream surface having a stepped configuration including at least two regions which each define a surface segment which is generally parallel to the end surface, at least some of the multi-stage flow control passageways extending through the inward body between the upstream surface and the surface segment of each of the at least two regions; the multi-stage flow control passageways being formed to provide varying degrees of energy attenuating capacity, and arranged

within the inward body so that those multi-stage flow control passageways exposed to fluid flow upon the initial opening of the valve body provide greater energy attenuating capacity than those multi-stage flow control passageways progressively exposed to fluid flow as the valve body continues to transition to the fully open position.

11. The flow control element recited in claim 10, wherein the flow control element further comprises at least one opening collectively defined by the peripheral body and the inward body, the opening being arranged so that when the flow control element is disposed within the valve housing, the opening will be exposed to fluid flow subsequent to a majority of the multi-stage fluid control passageways being exposed to fluid flow as the valve body continues to transition to the fully open position.

12. The flow control element recited in claim 10, wherein each of the plurality of multi-stage flow control passageways is comprised of a plurality of segments, each segment being inclined 45 degrees relative to the flow axis when the flow control element is disposed within the valve housing.

13. The flow control element recited in claim 12, wherein each of the plurality of multi-stage flow control passageways includes a rectangular cross section in a plane perpendicular to the flow axis when the flow control element is disposed within the valve housing.

14. The flow control element recited in claim 12, wherein each of the plurality of multi-stage flow control passageways includes a first rectangular cross section in a first plane perpendicular to the flow axis when the flow control element is disposed within the valve housing, and a second rectangular cross section in a second plane perpendicular to the flow axis when the flow control element is disposed within the valve housing, the second plane being downstream from the first plane and the second rectangular cross section being larger than the first rectangular cross section.

15. The flow control element recited in claim 12, wherein each of the plurality of multi-stage flow control passageways extends along a passageway axis which is parallel to the flow axis when the flow control element is disposed within the valve housing to define a respective length, the lengths of at least two of the plurality of multi-stage flow control passageways being different.

16. The flow control element of claim 10, wherein the downstream surface includes three regions which each define of surface segment which is generally parallel to the end surface.

17. A fluid control valve comprising: a valve housing having a fluid inlet and a fluid outlet; a flow axis connecting the fluid inlet and the fluid outlet; a valve body disposed within the valve housing and transitional relative to the valve housing between a closed position and a fully open position, in the closed position, the valve body preventing fluid flow between the fluid inlet and the fluid outlet, the valve body being incrementally transitional from the closed position to the open position to incrementally increase the amount of fluid flow from the fluid inlet to the fluid outlet; and a flow control element positioned within the valve housing downstream of the valve body, the flow control element having: an orifice plate having a concave upstream surface, a plurality of passage inlets formed therein and at least one flow opening formed therein; a single-piece main body downstream of the orifice plate and having a plurality of multi-stage flow control passageways extending therethrough between opposed, generally planar first and second surfaces, wherein adjacent stages in each of the multi-stage flow control passageways are separated by a turn or bend, the main body being aligned with the orifice plate such that at least some of the plurality of passage inlets are in communication with respective ones of the multi-stage flow control passageways; and a frame bisected by the flow axis and at least partially surrounding the main body such that a common axis extending perpendicularly from the flow axis extends through portions of the main body and the frame, the frame defining at least one opening in fluid communication with at least some of the plurality of passage inlets.

18. The fluid control valve recited in claim 17, wherein the main body includes an inlet face and an outlet face spaced from the inlet face along the flow axis, each of the plurality of multi-stage flow control passageways being comprised of a plurality of interconnected segments, each segment

being inclined 45 degrees relative to the flow axis.

19. The fluid control valve recited in claim 18, wherein each of the plurality of multi-stage flow control passageways extends along a passageway axis which is parallel to the flow axis to define a respective length, the lengths of at least two of the plurality of multi-stage flow control passageways being different.

20. A flow control element for use in a fluid control valve having a valve body disposed within a valve housing having a fluid inlet and a fluid outlet connected by a flow axis, and being transitional relative to the valve housing between a closed position and a fully open position, fluid flow through the fluid control valve incrementally increasing as the valve body incrementally transitions from the closed position toward the fully open position, the flow control element comprising: a peripheral body defining an end surface and bisected by the flow axis when the flow control element is disposed within the valve housing; and a single-piece inward body at least partially surrounded by the peripheral body such that when the flow control element is disposed within the valve housing, a common axis extending perpendicularly from the flow axis extends through portions of the inward body and the peripheral body, the inward body having a concave upstream surface which is complementary to an exterior profile of the valve body, a downstream surface opposite the upstream surface, and a plurality of multi-stage flow control passageways extending through the inward body between the upstream and downstream surfaces thereof, wherein adjacent stages in each of the multi-stage flow control passageways are separated by a turn or bend, the downstream surface having a convex configuration creating thickness variations in the inward body which in turn create corresponding length variations in at least some of the multi-stage flow control passageways; the multi-stage flow control passageways being formed to provide varying degrees of energy attenuating capacity, and arranged within the inward body so that those multi-stage flow control passageways exposed to fluid flow upon the initial opening of the valve body provide greater energy attenuating capacity than those multi-stage flow control passageways progressively exposed to fluid flow as the valve body continues to transition to the fully open position.

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